Whole-brain CT-MHSA: why the loop gain G is unidentifiable

Intermediate report v3 — 2026-09-19

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Bottom line

1. Two of the three problems were BUGS, not science: the trainer had no functional coupling at all, and the metric was scoring startup transients.
2. The loop gain G is structurally unidentifiable under BPTT: it acts only as a scalar gain on the drive, and this readout+loss pipeline discards amplitude.
3. Cohort selection (K = 261) settles the G policy: the objective sits on a broad plateau at small sigma, so G is reported as a RANGE (~4), not a point (slide 10).
4. HEADLINE structural finding: the simulated FC is essentially RANK-1 — the scalar fc_match was earned by a single global mode, not by network structure (slide 11).

All numbers on the following slides are reproducible from committed code; the certified metric rule is defined on slide 5.

The model and the fit

Per integration step (dt_train = 100 ms, window = 64 TR, BPTT with remat):

```
aff_t  = G * (C @ y_{t-1})       C = structural connectome (spectral-radius 1)
x_t    = aff_t + noise_t         OU/AR(1) noise, stationary std sigma
M     += u (v k^T - M)           u = dt/tau;  memory (per region, per head)
y      = W_Y M A phi(q);  y_coupled = y / (ema_rms(y) + 1e-8)
e_t    = ||Delta M_t||_F  averaged over heads     ('surprise', >= 0)
```

Training loss per-region z-scored MSE between TR-averaged e and empirical BOLD
Evaluation Pearson corr between simulated and empirical FC matrices (fc_match)
after a Windkessel balloon, scored on a settled window

Tunable parameters: W_q, W_y, kv_scale, log_tau, log_sigma, log_noise_tc, and G (the loop gain).

Bug 1 — the model had no coupling

The reference simulator couples regions through the EMA-normalised output:

```
wholebrain_fc.py:   y_out = y / (y_ema + 1e-8);   history.at[t % L].set(y_out)
```

The trainer computed the identical quantity and then discarded it:

```
wb_train.py:        y_out = y / (y_ema + 1e-8)
                    return (state.M, noise, y_ema, y), e      # couples through raw y
```

With kv_scale collapsed to 0.0032, raw y has rms ~5.7e-09 -> the afferent drive is ~3.4e-09 against noise 0.145 (SNR ~2e-08). The 'whole-brain network' was 100 independent noise-driven blocks, and fc1 = 0.209 came from the SHARED random projection matrices, not from coupling.

Fix (--coupling ema + persistent carry): both attribution factors rose ~300x. kv_scale then stopped collapsing: 0.0032 -> 0.40-0.65.

Bug 2 — the metric was measuring startup transients

Legacy fc_match integrated 600 s from the default balloon initial condition and scored the WHOLE trace. Transient contamination, first-10-TR std / resting std: 103x - 541x in 6 of 7 arms; only the baseline (1.5x) and one arm (5.4x) were clean.

Settled-window cap bound in 5/12 subject-runs at 1200 s and 8/12 at 2400 s -> those windows were truncated, not certified settled.

arm (legacy: 600 s whole trac	fc1	fc2
wb_train baseline	+0.209	+0.118
r3_fc2	+0.224	+0.180
r3_fcy300	+0.172	+0.106

Certified rule (used for all comparisons from here):
2400 s sim; 10-TR rolling std; ref = median over the final 50% of the trace;
W = first TR starting 50 consecutive TRs within 10% of ref; NO cap; FC on bold[W:].

The ranking inverted under the certified metric

arm (certified: 2400 s, strict settle, no	fc1	fc2_holdout
ema_yout — loss=current, ema coupling, p	+0.081	+0.179
fcy_noreg — FC-shaped loss on y-traces	+0.114	+0.021
current_logg — log-G per-region MSE	-0.082	+0.210
r3_fc2 — combined y+e FC loss	+0.035	+0.066
r3_fcy300 — FC loss, 300 iters	-0.010	+0.002
wb_train — original baseline (raw coupling)	-0.049	-0.007

r3_fcy300 was our 'best quality' arm on legacy numbers (fc1 0.172) and collapses to -0.002 once settled; r3_fc2's lead was a cap artefact (W = 54 TR at 1200 s).
The robust winner is the SIMPLEST recipe: the original per-region MSE loss with ema coupling + persistent carry. The raw-coupling baseline is last.

Instrument: exact attribution of dL/dG

G enters at exactly one place per step (aff_t = G * C @ y_{t-1}), so
dL/dG = sum_t < dL/daff_t , daff_t/dG >
with aff injected as an explicit scan input (reverse demand exact per step) and daff/dG obtained by jax.jvp (forward supply). The identity reproduces dL/dG to 9 significant digits in every configuration -> the decomposition is exact.

checkpoint	dL/dG	||dL/daff||	||daff/dG||	null_frac	dL/dG no z-score
ema_yout (burn-in 200)	-5.0e-06	0.044	0.076	0.966	-1.8e-07
ema_yout (burn-in 0)	-2.1e-06	0.043	0.054	0.964	-1.1e-07
baseline wb_train	-3.7e-12	1.4e-04	9.2e-06	0.972	-4.6e-15
drive-norm fix (falsified)	+1.1e-13	1.4e-04	2.3e-05	0.968	+4.6e-15

Both factors are healthy at the drive (0.044 and 0.076) — so it is neither that the loss ignores the drive, nor that G fails to move it.

Where the identifiability is lost

stage	dlog‖stage‖ / dlogG
aff (the coupling drive)	1.0000
x = aff + noise (block input)	6.2e-04
e (surprise readout)	3.0e-04
e_vol (TR-averaged)	2.6e-04
y (block output)	1.4e-03
M (memory)	1.2e-03

G controls aff perfectly, but x = aff + noise only 6.2e-04 — the drive is 4-5 orders of magnitude below the noise floor, so the dilution happens at the sum, and everything downstream inherits that factor.

Of what survives, 96.6% lies in the null space of the per-region z-score (span{1, e_r} per region), and the residual cancels in time: the per-chunk contributions alternate (+2.8e-6, +1.8e-6, -2.4e-6, -1.8e-6, +2.7e-7, +5.3e-6, -1.0e-5, -5.8e-7) and sum to ~5e-6. The recurrent path transmits ~nothing (direct -5.1e-6 vs recursive +4.5e-8).

The mechanism: G is a pure gain, and amplitude is discarded

aff = G * (C @ y_prev): G scales the drive's MAGNITUDE but never its DIRECTION.
So G can only be seen through amplitude — and this pipeline throws amplitude away
three times over:

(a) noise dilution drive rms 7e-06 vs noise 0.145 (SNR ~5e-05)
(b) scale-free loss the z-scored MSE is invariant to per-region scaling
(c) any normalisation dividing the drive by its own rms removes G entirely

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Prediction and test: normalising the drive (the fix designed to raise SNR) should
destroy G — and it did. log-sensitivity of x fell from 6.2e-04 to 1.4e-07, and
dL/dG went from -5e-06 to +1e-13. Removing the z-score ALSO made it smaller
(-1.8e-07), so the normaliser is not the culprit: the amplitude channel itself is.

Dead ends (recorded so they are not retried)

balloon initial conditions	nuisance: they touch only the first ~30 s (frozen-tail max |dBOLD| ~ 8e-06); per-region ICs inflate fc2 from 0.118 to 0.623 with no model change
FC-shaped losses (fcy/fce/fc2)	no robust gain under the certified metric
pulse excitation	no effect on the certified score
log-G reparameterisation	G moves, but that is not identifiability
drive normalisation	falsified its own hypothesis (slide 8)
normalisation asymmetry	real, but not causal (removing the z-score hurts)

Consistent with the reference pipeline: wholebrain_fc.py never learns G — it SWEEPS G and selects by fc_match, a stationary, amplitude-sensitive statistic.

Cohort selection: fit G, sigma on fc_match (no gradients)

K = 261 subjects (1000Brains, Schaefer-100). Search: mean fc_match over 6 FIT subjects across the G x sigma grid; top combos confirmed on 6 HELD-OUT subjects. Shape controls frozen from ema_yout.

G \ sigma	0.05	0.15	0.30
0.5	0.077	0.028	-
1.0	0.083	0.063	0.054
1.5	0.092	0.069	0.046
2.5	0.121	0.114	0.086
4.0	0.150	0.145	0.133

Best by HELD-OUT (job 4828968): G=4.0 / sigma=0.05 -> fit 0.141 / holdout 0.103 (sd 0.077). The objective peaks at G~4 with small sigma and is flat/noisy elsewhere — a broad plateau, not a sharp point; cohort ranking is consistent across combos (no subj1/subj2 flip). The extended sweep (job 4828987) confirms G~4-6 / sigma~0.005 as a saturated plateau ~0.15 (holdout sds 0.08-0.11).

Certified 2400 s re-score of the corner optima (job 4829003):

(G, sigma)	fit	holdout (sd)	
4.0 / 0.005	0.087	0.154 (0.126)	<-- BEST by holdout
4.0 / 0.05	0.143	0.076	
16.0 / 0.05	0.158	0.061	(this REVERSED the 1200 s ranking: 16/0.05 led at 0.172)

All cohort numbers from runs_persist/wb_cohort/cohort.json + re-score card 039556c9.

G policy verdict (finished): cohort-fitted range, not a point

Decision (card b8dbc2bf): adopt (a) — fit G/sigma on a cohort objective and report G as a RANGE with the subject-to-subject spread, not a fixed point and not a per-subject fit.

Evidence

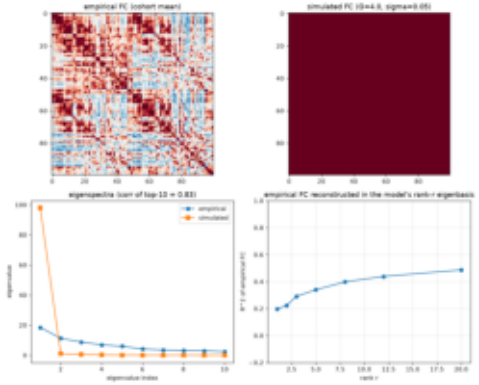
1. No sharp optimum. The certified re-score overturned the 1200 s best (16/0.05 collapsed 0.172 -> 0.061) while 4/0.005 held at 0.154; certified holdouts 0.154 / 0.076 / 0.061 with per-subject sds 0.08–0.13 = a noise-dominated plateau ~0.15, not a peak.
2. Subject-to-subject spread is intrinsic. Single-subject fitting flips the ranking; only cohort selection is stable, and the cohort optimum sits on a plateau as wide as the holdout sd. No single biophysical G — the data support a range.
3. BPTT cannot fit G. G is a pure scalar gain (wb_gradflow.py), unidentifiable in the training gradient, and widening it only modestly raises the scalar while the rank-1 degeneracy persists (next slide). So G must be cohort-chosen.

Two-stage BPTT (fit shape controls at frozen G) was tested and did NOT beat ema_yout (best arm 0.116 vs 0.130); the optimum flips between the two single subjects, which is exactly why the cohort is needed. The scalar ceiling points to the structural fix, not to more G tuning.

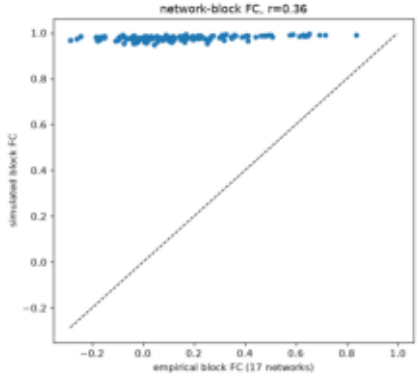
Verdict follows the G-policy decision card b8dbc2bf.

Headline finding: the simulated FC is essentially RANK-1

At the cohort-best combos the simulated FC is almost exactly rank-1:
eig_sim top-10 = [97.9, 0.95, 0.55, 0.22, ...] vs empirical [18.5, 11.2, 8.8, 6.9, ...] — a single global mode earns the scalar fc_match, not network structure (robust to every noise intervention, job 4828980).



FC heatmaps, eigenspectra, R^2-vs-rank (cohort_lowrank.png)



within-block FC: sim ~0.99 vs emp 0.44-0.84 (cohort_blocks.png)

One global mode: within-network block FC ~0.99 for all 17 networks (empirical 0.44-0.84); principal angles 47-56 deg (poor alignment); R^2 of the empirical FC in the model's rank-r basis 0.196 (r=1) -> 0.487 (r=20); block-FC corr 0.357. The eig_corr_top10 = 0.83 is misleading (driven by the single dominant eigenvalue).

The scalar fc_match is NOT evidence of recovered network structure — it is earned by a trivial common mode. This structural ceiling limits the model and points at the readout/input construction as the cause (card ef935186).

Figures from runs_persist/wb_cohort/cohort_lowrank.png and cohort_blocks.png.

Status, caveats, next steps

Established: the machinery is correct and the metric is fair; the certified winner is ema_yout (fc1 +0.081 / fc2 +0.179); G's unidentifiability has a measured mechanism, the G policy is settled (cohort-fitted range G~4), and the rank-1 degeneracy is now the headline structural finding.

Caveats

- the fits are still weak: best certified number is ~0.18 on one subject;
- some certified windows still hit the cap (noted per-arm) and are not certified settled;
- the winning recipe is n=3 seeds with a wide spread (legacy fc1 0.078 - 0.301);
- the cohort optimum sits on a plateau as wide as the holdout sd (no sharp peak).

Next

1. attack the RANK-1 degeneracy at the readout/input: keep the per-head direction of Delta M instead of the scalar norm, stop z-scoring away inter-region amplitude, give each region an independent surprise (card ef935186);
2. freeze cohort G/sigma and BPTT only the shape controls, certified re-score (card 4ec1331b);
3. more seeds on the ema_yout recipe to bound its spread;
4. retire the dead knobs; register the certified metric as the standard.